

CFES

CANADIAN FEDERATION OF ENGINEERING STUDENTS



FCÉG

LA FÉDÉRATION CANADIENNE DES ÉTUDIANT.E.S EN GÉNIE

Canadian Federation of Engineering Students

Document of Stances

English Edition

Revision Date: 20/11/2025
Revised By: CFES Board of Directors

Contents

Preamble	3
Mission	3
The CFES Advocacy Portfolio in Brief	3
The Document of Stances	3
Nomenclature	4
Engineering Accreditation	5
Language Electives	8
Mental Health and Student Workload	10
Quality of Engineering Internships	13
Sustainability	15
Equity, Diversity, and Inclusion	17
The Iron Ring Ceremony	21
Indigenous Education and Practices	24
Bilingualism of Canadian Official Languages	27
Online Accessibility and Academic Accommodations	30
Experiential Learning in Engineering Education	32

Preamble

Mission

One of the core mandates of the Canadian Federation of Engineering Students (CFES) is to advocate to relevant stakeholders on behalf of over 80,000 engineering students across Canada in 45 higher education institutions (HEIs). Over the years, the Federation has developed partnerships with organizations such as the Canadian Engineering Accreditation Board (CEAB) and Engineering Deans Canada (EDC) that have empowered the CFES to advocate on behalf of engineering students, influence decision making regarding the engineering curriculum and shape the future of the engineering profession.

The CFES Advocacy Portfolio in Brief

There are two key activities that inform the CFES' advocacy: the national stances and national surveys. The stances are intended to be updated regularly to reflect advances and setbacks in engineering student advocacy and are largely based on data collected from national student surveys which are intended to be released once a year.

The CFES has released three national surveys to date: 2017, 2020 on COVID-19-specific themes and 2023 based around student development and support. The CFES passed its first stances during a General Assembly at the Canadian Engineering Leadership Conference (then known as Congress) 2018, and have since been updated in 2024.

The Document of Stances

The Document of Stances summarizes the CFES' official member-approved stances to outline advocacy efforts in various aspects of the engineering student experience. Both the Document of Stances and the individual full-length stances are available publicly to members with their own advocacy efforts. These are inspired by Engineers Canada's position papers and serve as a valuable resource for members to inform institutional advocacy initiatives.

Nomenclature

ACES	Atlantic Council of Engineering Students
AWG	Advocacy Working Group
CEAB	Canadian Engineering Accreditation Board
CEWIL	Co-operative Education and Work-Integrated Learning Canada
CFES	Canadian Federation of Engineering Students
EngSoc	Engineering Student Societies
ESSCO	Engineering Student Societies' Council of Ontario
EDC	Engineering Deans Canada (<i>Note: formerly referred to as NCDEAS – National Council of Deans of Engineering & Applied Sciences</i>)
QCESO	Quebec Confederation for Engineering Student Outreach (<i>Note: the French acronym CREIQ is more commonly used</i>)
DC	Data Commissioner
SIWG	Social Issues Working Group
SU/s	Student Union/s
VPA	Vice President Academic
WESST	Western Engineering Student Societies' Team

Engineering Accreditation

Adopted:

CFES Congress 2018 [2018-01-07]

Last Updated:

2024-01-02

The Official Stance:

The Canadian Federation of Engineering Students believes that the accreditation system for Canadian engineering programs should protect the interests of engineering students by ensuring high, consistent and achievable standards for the quality of engineering education by involving student voices in the process of accreditation visits and the development of accreditation criteria.

The Students' Position

- Students are a key stakeholder in engineering education, and should be actively consulted in discussions and decisions related to engineering accreditation.
- The accreditation visit process should be reformed to better evaluate and incorporate student perspectives throughout the entire visit.
- An effective accreditation system should be primarily focused on learning outcomes, and any measure of student input should be representative of total learning time, and not favor lecture-based learning over other learning methods.
- Accreditation Units (AU) do not always favor engineering students, nor affect the quality of engineering education, where a more holistic approach should be taken either through AU or alternatively a different metric for completion of the engineering attributes.

Recommendations to Partners, Stakeholders, and Other Entities

- The CFES calls on the Canadian Engineering Accreditation Board (CEAB) to update its accreditation visit practices to incorporate more valuable and active student feedback at the end of each accreditation year.
- The CFES calls on the CEAB to continue building on efforts such as the Graduate Attributes to better reflect learning outcomes and total learning time.
- The CFES calls on the CEAB to revise the Accreditation Units, as education and concepts more than just a quantifiable number or checkbox as they may not accurately depict student understanding.
- The CFES calls on the CEAB to include CFES student representatives as equal stakeholders in all relevant projects, committees, and task forces related to accreditation [achieved].

- The CFES calls on the Engineering Deans Canada (EDC) to collaborate on joint recommendations to the current accreditation system based on the shared interests of faculty and students.
- The CFES calls on the transparent communication of EDC with their respective schools to ensure student voices can be heard at the time of the visit to ensure the best engineering education is provided.
- The CFES calls on the CEAB to consider incorporating more possibilities for extracurricular learning experiences into the current accreditation system. For the quality of such learning experiences, CFES will work with the CEAB and provisional engineering regulators to create a rigorous framework and set of standards.
- The CFES calls on the CEAB to standardize and regulate the quality of internships/coops offered by university engineering programs in Canada.
- The CFES calls on EDC to regularly update their course offerings and policies to match that of the changing AU requirements and improvements every academic year.

What the CFES is doing

- The CFES has provided student representation as an observer of the Canadian Engineering Accreditation Board (CEAB) for several years, and in 2015 formalized the responsibility to advocate to the CEAB.
- In 2017, the CFES sent representation to a meeting of the EDC for the first time, and has discussed future collaboration on issues related to accreditation.
- CEAB meetings grant the CFES speaking rights during open sessions as described by Engineers Canada Board Policy Below [1]:
 - 6.9.1 F.1 Observers at Meetings (1) The CEAB shall invite the following representatives to its meetings, as observers, each of whom shall be granted the right to be recognized as a speaker in the CEAB's open sessions: a) The president of the Canadian Federation of Engineering Students (CFES), or the CFES president's designate
 - In addition, Engineers Canada's Board Policy 7.2 directs the CEAB to maintain a relationship with CFES: c) Engineering students are a key stakeholder of accreditation. In addition to soliciting student feedback during program visits, the Accreditation Board is directed to maintain a relationship with the CFES and invite a representative to observe their meetings, requesting that they bring a report for the CEAB's consideration. All travel costs for this representative are covered by Engineers Canada.
- CEAB has updated its sample accreditation visit schedule based on feedback from stakeholders, including the CFES – creating more time for student engagement during the accreditation visits.

- The CFES has included an engineering curriculum section into the 2023 national survey to gauge the general knowledge of CEAB as well as understanding and inclusion of AU.
- Consultation with the CEAB was done for the 2023 national survey questions and updated stance documentation.

What the CFES plans to do

- The CFES will push for inclusion in all relevant projects and committees related to the CEAB.
- The CFES will continuously create and distribute a national survey in order to update its knowledge on student experiences with accreditation and accreditation-based services. Survey questions will be consulted with CEAB yearly to ensure no crossover with their own survey.
- The CFES will continuously update its stance on accreditation as significant changes in policy and regulations arise.
- The CFES will hold EDC accountable for untracked changes within their universities of new policy and regulations specific to engineering attributes.
- The CFES will gather additional feedback specifically related to accreditation visits to review the student perspective of the quality of the accreditation visits held.

Language Electives

Adopted:

CFES Congress 2018 [2018-01-07]

Last Updated:

2024-01-02

The Official Stance:

The Canadian Federation of Engineering Students believes that Canadian engineering students should have the opportunity to take elective language skills as part of their undergraduate program to enrich their engineering degree.

The Students' Position

- In 2018, the Canadian Engineering Accreditation Board (CEAB) updated their Accreditation Criteria & Procedures [2] to explicitly accept language electives as part of their “Complementary Studies” requirement for engineering programs across Canada; however, around 23% of accredited engineering schools in Canada have yet to reflect this development within their faculties.
- In 2022, the criterion was removed 3.4.5.2 as listed below, where the CFES has found it remains listed on certain complementary studies university websites:

REMOVED Criterion 3.4.5.2 Language instruction may be included within complementary studies provided it is not taken to fulfill an admission requirement. Furthermore, curriculum content that principally imparts language skills can be counted toward the required Accreditation Unit (AU) of complementary studies but cannot be used to satisfy the requirements for subject matter that deals with central issues, methodologies, and thought processes of the humanities and social sciences.

- CFES agrees with the removed statement, as even subject matter that deals with other languages is inclusive to the enrichment of engineering students apart from their technical courses.
- Knowledge of an additional language and surrounding subject matter is a valuable asset for the future of an engineering student and students must be given a full and appropriate range of language elective options.

Recommendations to Partners, Stakeholders, and Other Entities

- The CFES calls on the CEAB to assist the final 23% of Canadian institutions with the interpretation of their existing policy on complementary electives by publishing an official interpretive statement on the subject outlining additional criteria.
- The CFES calls on Engineering Deans Canada (EDC) to advocate for its member faculties to review and reconsider their elective offerings, and to allow engineering students to enroll in language courses that count towards their degree requirements, outside and including both Canadian official languages.

- The CFES calls on EDC to update their policies and information available on their academic calendars to accurately describe and follow the CEAB's Policies and Procedures as to what counts as a complementary course.
- The CFES calls on engineering student societies to support national advocacy efforts and leverage the CEAB policy update to advocate for the implementation of language electives in their program to reap the benefits of multilingualism in a diverse country and an increasingly globalized world.
- The CFES calls on their members to question the curriculum currently in place, and pursue studies that inspire them in language studies that would enrich their engineering education even if additional permissions are necessary to pursue.

What the CFES is doing

- The CFES' discussion on the interpretation of language course policy over two mandate periods was accepted by the CEAB's Policies and Procedures Committee who approved language sections as open electives before President's Meeting 2019.
- The CFES has created and continues to update an advocacy package to guide students in advocating for language electives at an institutional level.
- The CFES has studied the language elective options within accredited engineering universities in Canada, compiling a list of universities currently not following policy.
- The CFES has presented data on CEAB universities that do not currently follow policy, or have outdated information to begin that conversation between EDC and CEAB.

What the CFES plans to do

- The CFES will provide resources on language electives to equip its member schools with the information they require to advocate for elective changes at individual institutions.
- The CFES will provide an analysis on the benefits of having language electives available as complementary studies and how it enriches the education of an engineer.
- The CFES will work with EDC to make them aware of policies and documentation that go against or are not completely accurate based on current CEAB policy and procedure.
- The CFES will build on past advocacy efforts and continue the discussion on the importance of bilingualism/multilingualism during external partner presentations on the future of engineering education.

Mental Health and Student Workload

Adopted:

CFES Congress 2018 [2018-01-07]

Last Updated:

2024-01-02

The Official Stance:

The Canadian Federation of Engineering Students believes that engineering students face negative mental health outcomes as a consequence of the structure and workload demands of their programs. All stakeholders, including faculty, accreditors, and student leaders, have a responsibility to review and amend their practices to give students a healthy and effective learning environment.

The Students' Position

- Engineering students are at an elevated risk of negative mental health outcomes, with four themes remaining consistent throughout literature [3];
 - That engineering workload is a defining stressor.
 - Barriers exist that prevent engineering students from seeking help for mental health concerns.
 - Peers rely more heavily on one another to cope with stress and mental health distress.
 - The COVID-19 pandemic exacerbated student challenges surrounding mental health.
- Engineering students experience high levels of self-reported stress, anxiety and depression with some literature reportings of 28% moderate to extremely severe stress, 35% moderate to extremely severe anxiety and 34% moderate to extremely severe depression from a survey conducted in 2021 [4].
- The generalized perception that high stress and poor mental health is normalized and to be expected in engineering is harmful and dismissive of mental health issues for current undergraduate students and students wanting to choose engineering.
- Excessive student workload is likely a primary cause of these negative mental health outcomes, and steps should be taken to reduce academic stress related to workload.
- Poor mental health is disproportionately prevalent amongst women and other minorities including LGBTQ2S+ people, Indigenous people, and people of colour [4].
- Engineering students are unlikely to seek mental health support even if they self-report higher levels of mental health distress due to different factors including physical and cultural factors [5].

Recommendations to Partners, Stakeholders, and Other Entities

- The CFES calls on the Canadian Engineering Accreditation Board (CEAB) to investigate changes to accreditation criteria that better account for the full burden of student workload, and that require access to basic mental health services for all undergraduate engineering students.
- The CFES calls on the CEAB to investigate and review the Accreditation Units (AU) with respective hours to ensure they remain feasible within a full student workload.
- The CFES calls on the Engineering Deans Canada (EDC) to review the Mental Health Commission of Canada's National Standard for Mental Health and Well-Being for Post-Secondary Standards (MHCC Post Secondary Standards) [6] and implement these standards in their engineering faculties.
- The CFES calls on the EDC to hold their instructors accountable for ensuring workload does not harm engineering students in terms of mental health and instead work towards implementing more lifelong learning opportunities, problem based learning opportunities and less high stake academic challenges.
- The CFES calls on Engineers Canada and the EDC to partner with the CFES on research related to student mental health and workload, with the intention of clarifying the specific impacts of workload on the mental health of engineering students, and the best procedures for improving student mental health in engineering programs.
- The CFES calls on its partnered regional organizations (WESST, ESSCO, QCESO, ACES) to collaborate on future initiatives in student workload and mental health, potentially including a workload tracking system or a second national student survey.
- The CFES calls on the membership to evaluate their workload objectively and find gaps/problems within the engineering curriculum at their schools, and elevate the issue to their regional organizations.

What the CFES is doing

- The CFES has conducted a National Student Survey in 2023 to collect data on the mental health of engineering students as well as student workload/perception. The CFES has run sessions and talks on student mental health, with a focus on student initiatives and student volunteer burnout, at its major events including the Canadian Engineering Leadership Conference and the Conference on Diversity in Engineering.
- The CFES has created a bank of Mental Health Best Practices, available to all member societies via the external drive.

What the CFES plans to do

- The CFES will continue to facilitate the sharing of best practices related to mental health between its member societies, and update resources for student mental health advocacy when appropriate.
- The CFES will pursue further research into the specific consequences of negative student mental health, and best practices for improving mental health at engineering schools.
- The CFES will evaluate how it can better promote student mental health internally through the operations and content of its events.
- The CFES will follow up with relevant stakeholders including EDC to ensure systematic review is being done regularly of engineering curriculum workload, hours and policies in place to protect students.
- The CFES will review and apply MHCC Post Secondary Standards to conference planning, motivation for advocacy, and internal team practices.

Quality of Engineering Internships

Adopted:

CFES Congress 2018 [2018-01-07]

Last Updated:

2024-01-02

The Official Stance:

The Canadian Federation of Engineering Students believes that the quality of engineering internship programs across Canada is inconsistent and often substandard, and that engineering programs have a responsibility to revise their practices in order to offer students a better value for their program fees.

The Students' Position

- Student satisfaction with internship program quality is low across Canada, representing general dissatisfaction with the value of internship services offered at engineering schools including opportunities, resources provided and accessibility.
- The lack of regulation across engineering internship programs leads to greater variation and lower standards for the services that are offered to students, compared to other areas of engineering education.
- The flaws in internship programs vary considerably across institutions, meaning that an effective response to internship program issues must occur on a localized basis.

Recommendations to Partners, Stakeholders, and Other Entities

- Student satisfaction with internship program quality is low across Canada, representing general dissatisfaction with the value of internship services offered at engineering schools including opportunities, resources provided and accessibility.
- The lack of regulation across engineering internship programs leads to greater variation and lower standards for the services that are offered to students, compared to other areas of engineering education.
- The flaws in internship programs vary considerably across institutions, meaning that an effective response to internship program issues must occur on a localized basis.

What the CFES is doing

- The CFES included internship quality as a focus area for its 2017 National Student Survey and its associated research, in order to determine the scope of internship program issues across Canada.

- The CFES included internship and professional development questions in the 2023 National Survey to determine the requirements surrounding internship and resources available from institutions.

What the CFES plans to do

- The CFES will work with its partners to create resources for sharing basic information about engineering program practices across Canada, and for advocating to faculties for effective changes.
- The CFES will investigate the merits of advocating to Co-operative Education and Work-Integrated Learning Canada to improve their accreditation of mandatory engineering internship programs.
- The CFES will bring back data collected by the National Survey regarding internships to Engineering Deans Canada in order to provide direction to their individual programs.
- The CFES will work with the regional organizations (WESST, ESSCO, QCESO, ACES) to gain information at a regional level of common engineering internship practice and policy.

Sustainability

Adopted:

CFES Congress 2018 [2018-01-07]

Last Updated:

2024-01-02

The Official Stance:

The Canadian Federation of Engineering Students believes that sustainability is an essential consideration in engineering practice, and believes that it is necessary to educate and engage its members on issues of sustainability, while also evaluating and improving the sustainability in engineering curriculum.

The Students' Position

- Engineers have a responsibility to hold paramount the safety, health and welfare of the public and have regard for the environment, which entails a responsibility to perform work that is sustainable for future generations.
- The future generations of engineers should be thoroughly educated and invested in sustainable practices of engineering prior to graduation, through proper applications within the engineering curriculum.
- The CFES should evaluate the sustainability of its operations to reform and improve its practices as necessary.
- Advocacy work surrounding sustainability should be assessed within the lens of engineering students, job opportunities and the future of engineering.

Recommendations to Partners, Stakeholders, and Other Entities

- The CFES calls on the Canadian Engineering Accreditation Board (CEAB) to include sustainability-focused Accreditation Units (AUs) for all engineering degrees, and to ensure that every engineering student is graduating with those competencies.
- The CFES calls on members to perform institution-specific level advocacy focusing on including sustainability considerations in their everyday operations and in travel to respective CFES conferences (i.e. encouraging institutions to divest and provide sustainability resources).
- The CFES calls on Engineering Deans Canada (EDC) to support sustainability initiatives, teams and practices within their individual institutions and partner with organizations to create a more sustainable engineering campus.

What the CFES is doing

- The CFES hosts a national conference on sustainability in engineering annually at one of its member schools with the goal of directly educating and engaging engineering students on sustainable practices related to the profession that they can bring back and implement ideas into their universities.
- The CFES will continue to update the best practices for sustainability for use by both itself and its member societies Sustainability Bank of Best Practices, as well as other useful learning resources related to sustainability.

What the CFES plans to do

- The CFES will perform an annual environmental footprint analysis of its operations, and reform its operations to align with sustainable practices wherever possible.
- The CFES will assist to develop documentation for a conference guideline on sustainability to consider sustainable practices in their planning and execution processes. As a national student organization, the CFES has a major role to play in paving the way for sustainable actions as an organization and then further motivating the member schools.
- The CFES can develop guiding material for itself and member organizations to decrease their environmental impact, through a living document outlining best practices. This document shall be reviewed and updated annually to reflect any changes in best practices.

Equity, Diversity, and Inclusion

Adopted:

CFES Congress 2019 [2019-01-04]

Last Updated:

2024-01-02

The Official Stance:

The Canadian Federation of Engineering Students (CFES) believes that equity, diversity and inclusion (EDI) are essential to the sustainability of the engineering profession and its contribution to the public interest. Increased inclusion and participation of underrepresented groups bring different perspectives and experiences that help engineers when addressing problems within the engineering profession and throughout undergraduate curricula.

The Students' Position

- Increasing the diversity and minority group representation within engineering helps to tackle challenges and create solutions to engineering problems that benefit everyone.
- Engineering students should have access to equal opportunities in engineering regardless of their age, ethnicity, race, gender, sexual orientation, economic status, disability, religion, or other diverse backgrounds. These opportunities include access to resources and opportunities that further their growth as an engineer (i.e. internships, engineering conferences, etc.).
- Increasing the diversity of students in engineering programs creates an enhanced engineering education by exposing students to a more significant amount of perspectives, experiences, and ways of thinking.
- Diversity will be fostered at a higher and more successful rate when in an inclusive environment by contributing to the retention of underrepresented groups. In order to create an inclusive environment, education and initiatives must be created around topics of diversity to educate individuals about increasing accessibility with diverse peoples and reduce unconscious bias.
- Indigenous inclusion in engineering is an important, yet under-discussed topic given that Canada is on traditional and unceded Indigenous lands and exists as a result of colonization; the effects of which are still present today.
- The CFES further endorses the Engineers Canada National Position Statement on Diversity and Inclusion, and the Engineers Canada 30 by 30 initiative on raising the percentage of newly licensed engineers who identify as women to 30 percent by the year 2030.
- Underrepresented groups are often at a disadvantage due to systemic barriers that disproportionately impact them, which can make it difficult to enter engineering and thrive.

- Creating an inclusive environment within engineering means to create a safe environment, culture and practice that welcomes and respects the contributions of individuals from diverse backgrounds, experiences and perspectives.

Recommendations to Partners, Stakeholders, and Other Entities

- The CFES calls on the Engineers Canada “30 by 30 Network” to investigate the reasons why retention rates of women are low in undergraduate education and formally present to other stakeholders including the Engineers Canada Board of Directors.
- The CFES calls on Engineers Canada “30 by 30 Network” to begin developing secondary plans if 30 by 30 is not accomplished Canada wide, and integrate student voices in the new proposal.
- The CFES calls on Engineering Deans Canada (EDC) to continue promoting the recruitment of diverse identities within incoming engineering students with initiatives that promote equity within the engineering community. (i.e underrepresented community scholarships).
- The CFES calls on the Canadian Engineering Accreditation Board (CEAB) to integrate equity, diversity and inclusion (EDI) into the Graduate Attributes, similar to the impact of engineering on society and the environment, to describe the importance of EDI training and always ensuring EDI considerations are done in the engineering profession.
- The CFES calls on EDC to integrate an Associate Dean of Equity, Diversity and Inclusion or a position of similar standing within their institutions to achieve equitable, diverse and inclusive accessibility to engineering and provide support to underrepresented groups.
- The CFES calls on EDC to provide equity, diversity and inclusion training to their instructors and students, specifically students that represent the institution at large.
- The CFES calls on engineering faculties across Canada to hire diverse instructors that reflect the diversity of students they teach and ensure support for marginalized faculty.
- The CFES calls upon Institution-specific Administrators, with the support of Engineering Deans Canada to implement and enforce protocols to handle cases of discrimination and harassment within the engineering community.
- The CFES calls upon the CEAB and EDC to promote stewardship principles in curriculum, to ensure that students are better equipped to consider and champion EDI values in the workplace upon completion of their studies.

- Lastly, the CFES calls on our members to examine the engineering traditions that they uphold and consider with a critical lens whether or not they uphold and contribute to creating an inclusive environment for all. Furthermore, the CFES calls on members to seek out opportunities to promote EDI literacy amongst students, and create initiatives that support equity, diversity and inclusion in their institutions.

What the CFES plans to do

The CFES plans to strengthen our support and promotion of inclusion within the engineering student community by providing resources and sharing best practices to member schools regarding hosting inclusive events and developing inclusivity orientations/training within their respective communities.

- Continue its inclusivity efforts by offering inclusivity orientations at all events, as well as further investigating event inclusivity risk assessment.
- Explore new partnerships with diversity and inclusion-focused organizations (Catalyst Canada, Actua, Women in Engineering, Canadian Mental Health Association, National Society of Black Engineers, Diversity Initiative at the Engineering Change Lab, etc).
- Prioritise the active incorporation of bilingualism within all CFES Activities, as well as within the core of the organization.
- Develop inclusive event planning guidelines specific to CFES activities to help identify, remove, and prevent barriers that may hinder anyone from fully participating in CFES Activities and ensure these considerations are taken from the early planning stages of each activity.
- Fulfill all the commitments set out in the Champions Charter to officially endorse Engineer's Canada's 30 by 30 initiative, including nominating a Champion at Spring Meeting each year that participates in the Champions' group and actively engages in the 30 by 30 network.
- Put into practise the Exclusionary & Harmful Reporting Form that provides a space for feedback from members.
- Prioritize barrier-free events and activities at CFES conferences, with alternatives made available when this is not possible.
- Create a CFES Equity, Diversity and Inclusion Toolkit for members, that includes but is not limited to the following:
 - Equity, diversity and inclusion training
 - EDI certification program
 - Best practices for external organizations
 - Canadian statistics of engineering demographic information

- Access to CFES partners for local advocacy
- Feedback mechanisms

The Iron Ring Ceremony

Adopted:
CFES CESS [2023-03-02]

Last Updated:
2024-01-02

The Official Stance:

The Canadian Federation of Engineering Students believes that the Iron Ring Ceremony is a necessary transitional ceremony for engineering students, but in its current state is outdated and does not fully represent modern age engineering. It is instead limiting the description of the engineering profession and exclusionary in tone, and should be adjusted to affirm all engineers in their Obligation.

The Students' Position

- The Iron Ring Ceremony is an essential rite of passage for engineering students, placed in high regard for graduates after the completion of a long and difficult degree.
- At its best, the Iron Ring Ceremony can contribute to a sense of belonging and community in engineering.
- The original ritual is not inclusive to all engineering students, leaving them disappointed. The ritual restricts the practice of engineering outside of the original forms of work when the ritual was first enacted as well as presents an outdated worldview regarding colonialism, racism and sexism.
- Each camp conducts its own ceremony with little oversight from the corporation of seven wardens.
- All engineering students should feel included, comfortable and respected during The Ritual of the Calling of the Engineer.

The Issue

The Iron Ring Ceremony is the culmination of years of hard work for engineering students in Canada, a ritual where the student is called to take an obligation to practice engineering by ethical and professional standards [7]. This causes the ceremony to hold great importance to many undergraduate students. The main function of the ceremony is to make engineering students feel positive and affirmed as obligated engineers.

Currently, some undergraduate students leave the ceremony feeling a lack of accomplishment and belonging in the engineering profession, straying from the original purpose. Apart from student voices claiming discomfort and discontent within the current Ritual of the Calling of the Engineer, both engineers and engineering educators united to create a statement submitted to the Council of the Seven Wardens in 2022 with a movement “Retool the Ring” mainly outlining that it fails to embody a comprehensive understanding of engineering ethics, with a lack of clarity and transparency in text, holding outdated and harmful worldviews rooted in colonialism, racism and sexism [8]. They

sought for the Council to acknowledge and take responsibility in order to create a new Call of the Engineer that encompasses modern engineering practice.

The largest challenge with the Iron Ring Ceremony is that these issues do not exist everywhere as there are various currently standing forms of the poem, “The Ritual of the Calling of the Engineer”. Each Camp organizes the Iron Ring Ceremony independently for a certain geographic area of the country. This camp is allowed to run its own version of the ceremony while some have resolved aforementioned issues. Some Camps have opened the ceremony to guests, others have begun to commission new writings to use in the ceremony.

The CFES believes that all engineering students should be called to be an engineer in the ritual “Calling of the engineer” and no student should feel disrespected, uncomfortable or diminished due to their gender, age, religion, race or any outside attribute that has no effect on the engineering practice. As a rite of passage, it must be a ritual that enlightens and encourages the engineers of tomorrow, to practice and grow within their engineering profession.

Recommendations to Partners, Stakeholders, and Other Entities

- The Partners, Stakeholders, and Other Entities are encouraged to collaborate with the CFES to ensure that the recommendations from the membership are considered, in delivery of the Iron Ring Ceremony.
- Integrate both current undergraduate and postgraduate student voices into the current panel enacted titled “Ritual Review Committee” [9] for the revision of the Iron Ring Ceremony, based on the Corporation of the Seven Wardens.
- For the membership to gather direct information from graduates of changes they would like to see either through regional organizations or their own engineering societies.
- The CFES calls Engineers Canada to push for information to be given to undergraduate students before they attend the ceremony.

What the CFES is doing

- Working to collaborate with member organizations to share knowledge and understanding about the Iron Ring Ceremony.
- Gathering feedback from graduates that have undergone the Iron Ring Ceremony.

What the CFES plans to do

- Working to ensure proper oversight is conducted over the camps from the Corporation of the Seven Wardens, ensuring that all engineering students are able to positively experience the Iron Ring Ceremony.

- To survey engineering students across all camps about their experience during the Iron Ring Ceremony.
- Workshops/presentations aimed at disseminating information.
- Survey for current students and recent grads about the iron ring ceremony.
- Compiling pre-existing surveys.
- The CFES shall commit to conducting a minimum of five educational presentations or workshops annually, aimed at disseminating comprehensive information about the Iron Ring Ceremony. These sessions will be targeted towards our member base, ensuring widespread understanding of the ceremony.
- The Federation will execute a detailed survey initiative, (date TBD) with the objective of gathering insights from a minimum number of participants (TBD), encompassing current students and recent graduates. This survey will specifically probe themes of inclusivity, representation, impact, and logistics, ensuring that the feedback is relevant, broad-based, and actionable. The findings of this survey will be reviewed annually to track progress and evolving student perceptions.

Indigenous Education and Practices

Adopted:
CELC 2024 [2024-01-05]

Last Updated:
2024-01-05

The Official Stance:

The Canadian Federation of Engineering Students acknowledges the harm done to Indigenous Peoples and recognizes that true reconciliation is an ongoing process. The role the engineer holds within reconciliation includes the rebuilding of our relationship with Indigenous Peoples. To incorporate Indigenous perspective into engineering problem-solving, engineering curricula must include Indigenous knowledge, as well as institutions must provide additional efforts to increase Indigenous representation in engineering.

The Issue

Indigenous issues, knowledge, and socially responsive reconciliation should be prioritized within engineering education. Indigenous Peoples have undergone hardships throughout Canadian history, including the dispossession of ancestral lands and being denied the opportunity to lead as a primary consultant for issues relevant to their heritage. An intersectional perspective is fundamental to the understanding and mutual respect that must be established in order to move towards positive change.

Decolonization strives to create a safe space for all Indigenous staff, faculty, and students at institutions; more importantly protects local Indigenous Peoples to instead promote collaboration and local teachings. Due to the lack of localized Indigenous integration, there exists a disparity between education on history and the traditional engineering curriculum. Localized Indigenous education teaches how to communicate effectively and respectfully with Indigenous stakeholders and includes when to engage and what issues need to be addressed specific to the geographic region.

Within the CFES, we believe Indigenous education and engineering should hold an emphasis on the interactions with Indigenous peoples. We believe that the Canadian Engineering Accreditation Board currently lacks localized Indigenous education as a Graduate Attribute, in order to observe an intersectional perspective and create well-rounded engineers that are able to translate Indigenous knowledge and communicate with Indigenous communities into their engineering profession. Changes must be made to educate individuals, including changing curriculum and having conversations over Indigenous practices and improvements within engineering education.

The Students' Position

- There is a lack of Indigenous representation in engineering when we consider only engineers in their main working years (age 25-64) with an education at a bachelor level or above, only 0.93% identify as Indigenous [10].
 - In 2021, 49.2% of Indigenous people in their working years had completed a postsecondary certificate, degree or diploma, a rate lower than that of non-Indigenous people at 68.0% [11]

- In 2021, Indigenous Peoples accounted for 5.0% of the total population in Canada, around 1.8 million Indigenous people [12]
- Increasing representation of Indigenous people in post secondary institutions should be a top priority with the assistance of government support for post-secondary education.
- Providing increased representation in the engineering profession can respond to the Truth and Reconciliation Commission's calls to action.
- Indigenous knowledge and wisdom must be a fundamental part of engineering education, to contribute perspective, traditional knowledge, and sustainable practices.
- Engineers interact with and directly impact Indigenous communities through the development, maintenance, and modification of infrastructure and engineering projects, therefore an intercultural understanding based in mutual respect must be found to promote positive communication and contribute to Reconciliation.

Recommendations to Partners, Stakeholders, and Other Entities

- The CFES calls on Engineers Canada to hear Indigenous engineering student voices for the improvement of representation of Indigenous Peoples.
- The CFES calls on Engineers Canada to involve the CFES in the DIEEN subnetwork to increase CFES knowledge for resources and surveys.
- The CFES calls the Canadian Engineering Accreditation Board (CEAB) to improve upon the current Graduate Attributes and expand to include local Indigenous Practices and Education to include communication with Indigenous communities and Indigenous knowledge and wisdom.
- The CFES calls on the Engineering Deans Canada to include Indigenous Education and Practices into strategic plans, to ensure institutions hold this practice in high regard when making decisions.
- The CFES calls on Engineering Deans Canada to integrate Indigenous Education and Practices into their institutions by way of activities, lectures and events following the example of other post-secondary institutions.
- The CFES calls on Engineering Deans Canada to promote the Indigenous recruitment within engineering, providing resources and making engineering accessible to Indigenous Peoples.

What the CFES is doing

- CFES conferences currently feature Indigenous Education and Practice through speakers, workshops, and different activities.

- The CFES is involved in the iBET PhD Project that supports Canadian Indigenous and Black professors in engineering and computer science fields across Canada.
- The CFES uses the National Survey to create a demographic landscape, including the identification of Indigenous peoples and participation in engineering.

What the CFES plans to do

- Connect with the Engineers Canada community of practice called Decolonizing and Indigenizing Engineering Education Network (DIEEN) to address Indigenous access to engineering education and truth and reconciliation within engineering education [13].
- Gather information from Engineering Deans Canada from institutions that have established post secondary engineering initiatives to promote Indigenous education to develop a framework for advocacy surrounding Indigenous Education (as per Engineers Canada documentation):
 - University of Alberta Indigenous Peoples and Technoscience Online Course
 - Indigenous Futures in Engineering at Queen's University
 - University of Manitoba's Engineering Access Program
 - University of Saskatchewan Indigenous Engineering Access Program
- Work with other partner organizations such as the American Indian Science and Engineering Society (AISES) to gain further insight and partner for further advocacy efforts.
- Analyze current information and surveys that have been collected by DIEEN and Engineers Canada to create an initial parameter for data collection and qualitative/quantitative analysis to then integrate questions about Indigenous Education into the National Survey.

Bilingualism of Canadian Official Languages

Adopted:

CELC 2024 [2024-01-05]

Last Updated:

PM 2024 [2024-09-22]

The Official Stance:

The Canadian Federation of Engineering Students supports Canada's two official languages; French and English as per the Canadian Charter of Rights and Freedoms. As a non-profit organization, the Canadian Federation of Engineering Students holds bilingualism at the highest priority. This is to reduce the systemic gap between the Francophone and Anglophone engineering student membership so that opportunities are not limited because of language barriers.

The Students' Position

- There exists a current language barrier between English and French speaking members.
- There is a lack of standardized procedures for fully bilingual CFES conferences and activities.
- There is a lack of bilingual universities across Canada, as well as a lack of resources to learn a fluent level of either official language within the engineering degree.
- The CFES believes all members should feel respected, confident, and heard when speaking or directing any CFES business in either language.
- Increasing bilingualism leads to a more inclusive space, with the opportunity to communicate more effectively amongst the membership.

Recommendations to Partners, Stakeholders, and Other Entities

- The CFES calls the Canadian Engineering Accreditation Board (CEAB) to review the language elective policy to provide the option to learn the second official language of Canada, if both are not known.
- The CFES calls the CEAB to work alongside Engineering Deans Canada (EDC), to find a place for bilingualism in all schools, regardless of the majority language spoken in an institution.
- The CFES calls EDC to support bilingual education and establish it as one of their strategic priorities going forwards in order to close the language gap and provide more opportunities to engineering students on a national level.
- The CFES will research partnerships for adult education services that provide French and English education opportunities (tutoring, courses, etc.) to CFES members at a reduced rate. The goal of this is to encourage optional extracurricular learning to bridge the language barrier between Francophone and Anglophone students.

- The CFES calls our partners to provide bilingual presentations to ensure that both anglophones and francophones can better understand and follow what is being presented.
- The CFES calls the activity managers and CFES officers to ensure all CFES activities remain safe spaces for Francophone minorities through the use of accurate translations and live interpretation.
- The CFES calls the members to seek out opportunities to promote bilingualism amongst students and create initiatives that support bilingualism in their institutions.

What the CFES is doing

- Utilizing live interpretation during General Assemblies and other official policy meetings within the CFES.
- Investigating the language elective policy within the Canadian Engineering Accreditation Board to see which universities restrict how many courses of the other official language can be taken.
- Utilizing current artificial intelligence tools to assist with translation online and in person.
- The CFES has a bilingualism commissioner, supervised under the VP Communications, as well as an adjoining Bilingualism Working Group to assist with translation of documents, presentations and policy.
- During the election of the CFES national executives, each question must be asked in both official languages by alternating which language is asked first. All candidates will be asked about their level of comprehension of both French and English as well as their plan to communicate with delegates whose first language does not match their own.

What the CFES plans to do

- Create a standardized procedure for translation and interpretation and planning specific procedures for CFES activities to ensure all activities and conferences are fully bilingual and a priority from the beginning.
- Provide support to CFES officers who are not fluent in both official languages with resources to help them in the language they are not comfortable in.
- Ensure safe spaces are created where both languages can be spoken freely with the support of interpretation when necessary.

- Prioritize bilingual presenters in events and activities at CFES conferences, with alternatives for unilingual presenters to be understood by the conference attendees.
- Put into practice a Bilingualism Practice Form that provides feedback from members.
- Explore partnership with translation services and bilingual focused organizations in the specialization of engineering.
- Put into place a system of formal and informal mentorship within the CFES to encourage practicing languages in a low stress environment.
- Create a CFES Bilingualism Toolkit for members, that includes but is not limited to the following:
 - Bilingualism training
 - Templates for dual presentation
 - Best practice for intercommunication
 - A list of words and expressions
 - Feedback mechanisms

Online Accessibility and Academic Accommodations

Adopted:

CELC 2024 [2024-01-05]

Last Updated:

2024-01-05

The Official Stance:

The Canadian Federation of Engineering Students believes that educational resources [recordings, lecture notes etc.] should remain accessible online at the same level of quality as in-person sessions. The CFES further believes that accommodations need to be in place for assessments throughout the academic term for extenuating circumstances, illness, and other personal matters, providing catered support to all engineering students.

The Students' Position

- All engineering students should have equal opportunity to pursue and excel in their education, regardless of individual abilities and circumstances.
- COVID-19 presented the opportunity to have permanent online resources that have been favored by engineering students to improve their academic performance.
- A universal design approach for educational resources and examinations does not accommodate for diverse needs and learning challenges.
- Traditional assessments do not reflect understanding of content in all situations due to the high stress environment and necessity to perform on the spot.
- Accommodations should be provided on a need basis to cater and support engineering students in their education and to encourage their success.

Recommendations to Partners, Stakeholders, and Other Entities

- The CFES calls on the Canadian Engineering Accreditation Board (CEAB) to encourage institutions in the process of updating their institution policies and being more inclusive to hybrid education and online permanent materials.
- The CFES calls on the Canadian Engineering Accreditation Board to assess the accommodations and accessibility available in an engineering program when undergoing an accreditation visit.
- The CFES calls on the Engineering Deans Canada to encourage institutions to create a standard hybrid model for course delivery based on student feedback on the program.
- The CFES calls on the Engineering Deans Canada to ensure accommodations are granted to those with diverse circumstances or disadvantages in order to promote equity in engineering.

What the CFES is doing

- The 2023 National Survey is collecting information on academic resources and availability.
- Literature reviews on education in a post-pandemic world that found positives from online learning.
- The CFES continues to meet with stakeholders surrounding the flexibility of education and quality of education.

What the CFES plans to do

- The CFES will work with stakeholders and partners to find an even ground to begin policy work for online accommodations and academic accessibility in engineering education.
- The CFES will create resources to support the membership to advocate for online accessibility and academic accommodations through letters of support.
- The CFES will gather more information and testimonials on current student policy and experiences surrounding online accessibility and academic accommodations.

Experiential Learning in Engineering Education

Adopted:

CFES 2024 Fall General Assembly [2024-10-27]

Last Updated:

2024-10-27

The Official Stance:

The Canadian Federation of Engineering Students (CFES) believes that engineering education should be hands-on and engaging to prepare future engineers for their upcoming careers. The Canadian Federation of Engineering Students further believes that it is essential to expose students to experiential learning through core courses throughout the degree and that institutions should be taking student input on experiential learning.

The Students' Position

- The level of experiential learning currently available to Canadian engineering students is heavily dependent on the school being attended, and the province in which the school is located.
- Although students can gain experiential learning through internships and co-ops, these opportunities are not guaranteed, nor all-encompassing. It would thus be beneficial to increase the availability of experiential learning offered throughout academic semesters by making coursework more practical and introducing designated environments in which students can develop hands-on skills.
- In 2023, the Canadian Engineering Accreditation Board (CEAB) updated their Accreditation Criteria & Procedures to add new criteria of 3.4.4.4 as listed below:
 - A minimum of 225 AU of engineering design curriculum content in an engineering program shall be delivered by faculty members holding professional engineering licensure as specified in the Interpretive statement on licensure expectations and requirements [CITE].
 - The CFES agrees with the new criteria added by the CEAB as these faculty members who hold professional engineering licensure will improve the availability and quality of experiential learning opportunities to students that will prepare students to enter the workforce upon graduation.

Recommendations to Partners, Stakeholders, and Other Entities

- The CFES calls on the CEAB to update its accreditation visit practices to incorporate and gather more feedback from students regarding experiential learning, including hands-on aspects of core courses.
- The CFES calls on CEAB to create more requirement specifications on the amount of experiential learning required for an engineering degree.

- The CFES calls on the Engineering Deans Canada (EDC) to advocate to its member faculties to better construct future curriculums to have a balance of theory and practical skills within the core courses that engineering students are required to take.
- The CFES calls on the EDC to direct its member faculties to provide various supports, such as student clubs, to promote experiential learning opportunities along with the faculty.
- The CFES further calls on the EDC to direct its member faculties to compile and summarize the wide range of experiential learning opportunities available to their students through a unified resource page, such as an Experiential Learning Hub. This initiative will further ensure that all engineering students can easily discover and access experiential learning opportunities to enrich their degrees.
- The CFES urges the EDC to direct its member faculties to take feedback from students regarding experiential learning opportunities.
- The CFES calls on its partnered regional organizations (WESST, ESSCO, QCESO, ACES) to collaborate on advocacy efforts to ensure a balance of theory and practical skills are emphasized in their respective regions.
- The CFES calls on engineering student societies to support national advocacy efforts and leverage the CEAB policy update to advocate for the implementation of experiential learning in their program.

What the CFES is doing

- The CFES currently offers the opportunity to take part in the Board of European Students of Technology (BEST) courses through partnership, which allows students to cover a technical topic that could be related to career skills.
- The CFES holds an annual Canadian Engineering Competition (CEC), that incorporates innovation, creativity, theory, and practical skills throughout its eight (8) competition streams: Re-engineering, Parliamentary Debate, Junior Design, Senior Design, Programming, Innovative Design, Engineering Consulting, and Engineering Communication.
- The CFES organizes case competitions at select conferences, providing students with a platform to refine and showcase their problem-solving skills and apply academic knowledge.

What the CFES plans to do

- The CFES will collect data on experiential learning through the national survey.

- The CFES will follow up with relevant stakeholders including EDC, to address curriculum changes to increase opportunities for experiential learning in Canadian engineering faculties.
- The CFES will emphasize the significance and need for experiential learning during the CEAB and Engineers Canada board meetings.
- The CFES will maintain ongoing communications with the International Federation of Engineering Education Societies (IFEES) regarding ways to better advocate for experiential learning opportunities and exchange programs offered to Canadian engineering students.
- The CFES will collaborate with the Canadian Engineering Education Association (CEEA) to improve the quality and quantity of experiential learning opportunities in member schools.